

Adverse childhood experiences moderate the association between sensory processing sensitivity and depression

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Methods

558 undergraduates ($M_{age} = 20.19$, $SD = 4.03$) completed an online survey in return for course credit. 69.9% identified as female, 28.3% identified as male, 1.8% identified as other. 92.3% identified as Hispanic. Participants were retained for analysis if they reported their age, identified as male/female, and completed the following measures ($N = 521$). SPS was measured using the 27-item **Highly Sensitive Person Scale (HSPS)**, depression severity was assessed using the 9-item **Patient Health Questionnaire (PHQ-9)**, and ACEs were quantified through the 10-item **Adverse Childhood Experiences Questionnaire (ACE-Q)**. Statistical analyses were performed using SPSS 28 with 5,000 bootstrapped samples. Moderation analysis was based on ordinary least squared regression, using Hayes' PROCESS Macro V4.2, controlling for age and sex.

Discussion

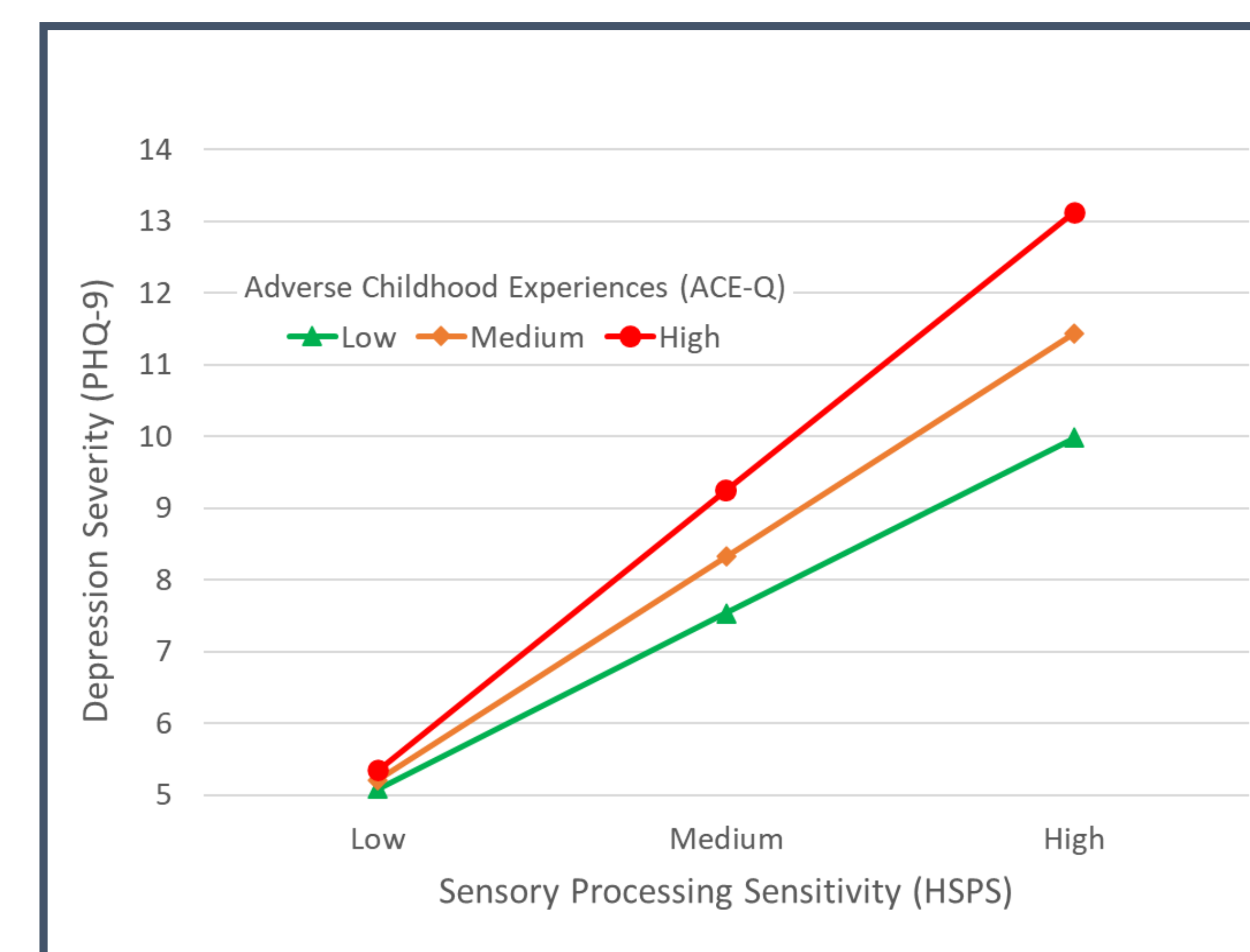
Our results support prior findings, showing that **greater SPS is associated with greater depression severity**. Importantly, the findings build on the limited existing evidence regarding moderators of this association. Specifically, **the data lends support to a diathesis-stress model in which the association between SPS and depression is amplified in adults who experienced more adverse events during childhood**. Recent research suggests that physical exercise may moderate the SPS-depression relationship (Yano & Oishi, 2018). Future research will need to identify modifiable factors such as this that could serve as potential intervention targets and to examine the extent to which ACEs might moderate the efficacy of such treatment approaches.

Introduction

More than one third of young adults report having been diagnosed with depression during their lifetime. Investigations of the link between personality and depression provide a path to identifying at-risk individuals and developing tailored treatment. One trait that has received increased attention in this regard is sensory processing sensitivity (SPS), a heritable trait associated with individual differences in processing of internal and external stimuli. SPS is therefore increasingly viewed as a diathesis for the development of depression. Furthermore, the diathesis-stress model predicts that this increased vulnerability is moderated by interindividual differences in stressful experiences, including adverse childhood experiences (ACEs). However, this theory has rarely been empirically examined. It is hypothesized that (i) SPS will be positively correlated with depression severity and (ii) that this association will be moderated by ACEs.

Results

65.3% of participants reported having experienced one or more ACEs, 15.2% reported four or more. 37.8% met the threshold for depression ($PHQ-9 \geq 10$). **SPS was positively correlated with depression** ($r(519) = .52$, $p < .001$). ACEs moderated the relationship, indicating that **the association between SPS and depression varied as a function of ACEs**. The interaction was probed by testing the conditional effects of SPS at three levels of ACEs, (-1SD, mean, +1SD). SPS was significantly related to depression at all three levels of ACEs. The overall model was statistically significant ($F(5, 515) = 47.49$, $p < 0.001$), predicting 32% of the variance in depression severity.



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